No.	Status	Date
LNDE-FN-20241025-031-DS-3-RV-101	Issued	2024-11-07

UT Thickness Data Sheet - 3-RV-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness (UT) measurement results for line 3-RV-101, located within the ATU-100. The scope of work involves the verification of wall thickness at designated condition monitoring locations (CMLs) to assess asset integrity and remaining life.

Marcus Chen, acting as the lead UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., performed the measurements and is responsible for the accuracy of the data recorded herein.

Technical Basis and Methodology

All thickness measurements were conducted in accordance with the requirements of ASTM E797. The field execution followed the specific site protocols defined in QSCP-NDT-001 Rev 3, ensuring alignment with the required CML-id formats and digital export procedures.

The survey was executed as part of the field work documented in LNDE-FN-20241025-031, which covers the UT thickness survey for 150 CMLs within the unit.

Component Design Data

The subject of this survey, 3-RV-101, is a regenerator overhead line constructed from A106 Gr.B. The line has a nominal outside diameter of 88.9 mm and a nominal wall thickness of 7.62 mm.

The design limits for this component are established at a maximum allowable pressure of 345 kPa(g) and a temperature of 121 °C. The minimum allowable thickness (t-min) for the line is 3.56 mm.

Thickness Measurement Results

UT readings for 3-RV-101 indicate several locations where the current wall thickness has fallen below the t-min of 3.56 mm. Specifically, CML 2-RF-102-04 was measured at 1.94 mm, 2-RF-102-05 at 1.92 mm, 2-RF-102-06 at 1.92 mm, and 2-RF-102-07 at 1.86 mm.

Additionally, CML 4-RA-101-06 was measured at 3.59 mm. While this reading is above the absolute t-min, it is noted as expired based on the calculated remaining life and the 3-year RBI interval for Class 1 sour gas regenerator overhead lines.

Integrity Assessment

Based on the site retirement thickness criteria, the readings at CMLs 2-RF-102-04, 2-RF-102-05, 2-RF-102-06, and 2-RF-102-07 are below the 3.56 mm threshold. These four locations are identified as expired and require the issuance of an NCR.

The remaining life for these locations is expired. Given the Class 1 designation of the regenerator overhead, the integrity of these sections is compromised relative to the design minimum thickness.

Thickness readings, measured 2024-11-04

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2022-09-16	5.82	5.64	0.052	0.057	36.2	2027-11-04	
6-SG-101-02	7.11	3.56	2022-09-16	5.42	5.12	0.067	0.077	20.2	2027-11-04	
6-SG-101-03	7.11	3.56	2022-09-16	5.29	5.25	0.064	0.061	26.5	2027-11-04	
6-SG-101-04	7.11	3.56	2022-09-16	5.19	4.98	0.074	0.071	19.3	2027-11-04	
6-SG-101-05	7.11	3.56	2022-09-16	4.72	4.65	0.087	0.077	12.5	2027-11-04	
6-SG-101-06	7.11	3.56	2022-09-16	5.55	5.39	0.059	0.034	31.2	2027-11-04	
6-SG-101-07	7.11	3.56	2022-09-16	6.16	6.04	0.040	0.040	>50	2027-11-04	
6-SG-101-08	7.11	3.56	2022-09-16	5.57	5.54	0.058	0.046	34.4	2027-11-04	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2022-09-16	4.77	4.62	0.088	0.085	12.1	2027-11-04	
6-SG-102-02	7.11	3.56	2022-09-16	4.64	4.59	0.088	0.101	10.2	2027-11-04	
6-SG-102-03	7.11	3.56	2022-09-16	4.69	4.45	0.090	0.093	9.6	2027-11-04	gauge photo attached
6-SG-102-04	7.11	3.56	2022-09-16	4.62	4.39	0.091	0.101	8.2	2027-11-04	gauge photo attached
6-SG-102-05	7.11	3.56	2022-09-16	5.96	5.89	0.043	0.036	>50	2027-11-04	
6-SG-102-06	7.11	3.56	2022-09-16	5.79	5.82	0.043	0.042	>50	2027-11-04	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	2019-10-18	7.63	7.57	0.022	0.012	>50	2030-11-04	
8-TG-101-02	8.18	4.06	2019-10-18	7.33	7.21	0.035	0.024	>50	2030-11-04	
8-TG-101-03	8.18	4.06	2019-10-18	7.37	7.27	0.033	0.020	>50	2030-11-04	
8-TG-101-04	8.18	4.06	2019-10-18	7.38	7.29	0.028	0.018	>50	2030-11-04	
8-TG-101-05	8.18	4.06	2019-10-18	7.57	7.54	0.024	0.006	>50	2030-11-04	
8-TG-101-06	8.18	4.06	2019-10-18	7.75	7.61	0.028	0.028	>50	2030-11-04	
8-TG-101-07	8.18	4.06	2019-10-18	7.17	6.98	0.047	0.038	>50	2030-11-04	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	2019-10-18	4.70	4.46	0.053	0.048	24.0	2030-11-04	
4-LA-101-02	6.02	3.18	2019-10-18	5.26	5.23	0.027	0.006	>50	2030-11-04	
4-LA-101-03	6.02	3.18	2019-10-18	5.38	5.19	0.027	0.038	>50	2030-11-04	
4-LA-101-04	6.02	3.18	2019-10-18	5.29	5.13	0.033	0.032	>50	2030-11-04	
4-LA-101-05	6.02	3.18	2019-10-18	5.38	5.25	0.027	0.026	>50	2030-11-04	
4-LA-101-06	6.02	3.18	2019-10-18	5.22	4.91	0.040	0.061	28.2	2030-11-04	
4-LA-101-07	6.02	3.18	2019-10-18	4.86	4.49	0.052	0.073	17.9	2030-11-04	
4-LA-101-08	6.02	3.18	2019-10-18	5.04	4.70	0.067	0.067	22.6	2030-11-04	
4-LA-101-09	6.02	3.18	2019-10-18	4.86	4.57	0.049	0.057	24.2	2030-11-04	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	2019-10-18	5.09	5.03	0.035	0.012	>50	2030-11-04	
4-LA-102-02	6.02	3.18	2019-10-18	5.15	5.08	0.035	0.014	>50	2030-11-04	
4-LA-102-03	6.02	3.18	2019-10-18	5.15	4.93	0.040	0.044	40.2	2030-11-04	
4-LA-102-04	6.02	3.18	2019-10-18	4.65	4.40	0.055	0.050	22.0	2030-11-04	
4-LA-102-05	6.02	3.18	2019-10-18	5.23	5.06	0.033	0.034	>50	2030-11-04	
4-LA-102-06	6.02	3.18	2019-10-18	5.52	5.42	0.020	0.020	>50	2030-11-04	
4-LA-102-07	6.02	3.18	2019-10-18	5.32	5.13	0.030	0.038	>50	2030-11-04	
4-LA-102-08	6.02	3.18	2019-10-18	5.55	5.38	0.037	0.034	>50	2030-11-04	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	2019-10-18	6.40	6.27	0.026	0.026	>50	2030-11-04	
6-LA-103-02	7.11	3.56	2019-10-18	6.49	6.28	0.032	0.042	>50	2030-11-04	
6-LA-103-03	7.11	3.56	2019-10-18	6.23	5.95	0.039	0.055	43.1	2030-11-04	
6-LA-103-04	7.11	3.56	2019-10-18	5.85	5.64	0.053	0.042	39.0	2030-11-04	
6-LA-103-05	7.11	3.56	2019-10-18	5.81	5.43	0.061	0.075	24.8	2030-11-04	
6-LA-103-06	7.11	3.56	2019-10-18	5.46	5.10	0.072	0.071	21.4	2030-11-04	
6-LA-103-07	7.11	3.56	2019-10-18	5.93	5.73	0.051	0.040	42.9	2030-11-04	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-08	7.11	3.56	2019-10-18	6.19	—	—	—	—	—	inaccessible – no reading
6-LA-103-09	7.11	3.56	2019-10-18	6.16	5.92	0.048	0.048	49.6	2030-11-04	
6-LA-103-10	7.11	3.56	2019-10-18	6.00	5.74	0.064	0.051	34.0	2030-11-04	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2022-09-16	6.83	6.50	0.070	0.156	17.2	2027-11-04	
4-RA-101-02	8.56	3.81	2022-09-16	6.79	6.46	0.073	0.184	14.4	2027-11-04	
4-RA-101-03	8.56	3.81	2022-09-16	5.98	5.52	0.108	0.222	7.7	2027-11-04	gauge photo attached
4-RA-101-04	8.56	3.81	2022-09-16	7.21	6.90	0.060	0.127	24.4	2027-11-04	
4-RA-101-05	8.56	3.81	2022-09-16	6.73	6.37	0.078	0.160	16.0	2027-11-04	
4-RA-101-06	8.56	3.81	2019-10-18	6.94	3.59	0.427	0.664	expired	immediate	< t-min; gauge photo attached
4-RA-101-07	8.56	3.81	2022-09-16	7.50	7.21	0.048	0.093	36.5	2027-11-04	
4-RA-101-08	8.56	3.81	2022-09-16	5.55	5.06	0.126	0.275	4.5	2027-02-10	gauge photo attached
4-RA-101-09	8.56	3.81	2022-09-16	7.37	7.17	0.049	0.101	33.3	2027-11-04	
4-RA-101-10	8.56	3.81	2022-09-16	7.19	6.91	0.061	0.127	24.5	2027-11-04	
4-RA-101-11	8.56	3.81	2022-09-16	6.07	5.53	0.110	0.224	7.7	2027-11-04	gauge photo attached
4-RA-101-12	8.56	3.81	2022-09-16	6.41	6.10	0.184	0.184	12.4	2027-11-04	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2022-09-16	7.42	7.33	0.042	0.050	>50	2027-11-04	
4-RA-102-02	8.56	3.81	2022-09-16	6.43	5.99	0.090	0.192	11.3	2027-11-04	
4-RA-102-03	8.56	3.81	2022-09-16	6.56	6.24	0.081	0.160	15.1	2027-11-04	
4-RA-102-04	8.56	3.81	2022-09-16	6.64	6.38	0.075	0.131	19.7	2027-11-04	
4-RA-102-05	8.56	3.81	2022-09-16	6.56	6.23	0.081	0.164	14.7	2027-11-04	
4-RA-102-06	8.56	3.81	2022-09-16	6.65	6.35	0.075	0.158	16.0	2027-11-04	
4-RA-102-07	8.56	3.81	2022-09-16	6.85	6.53	0.072	0.153	17.8	2027-11-04	
4-RA-102-08	8.56	3.81	2022-09-16	5.86	5.26	0.113	0.226	6.4	2027-11-04	gauge photo attached
4-RA-102-09	8.56	3.81	2022-09-16	6.42	6.04	0.088	0.147	15.2	2027-11-04	
4-RA-102-10	8.56	3.81	2022-09-16	6.86	6.56	0.158	0.158	17.4	2027-11-04	
4-RA-102-11	8.56	3.81	2022-09-16	7.33	7.19	0.073	0.105	32.2	2027-11-04	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2022-09-16	9.55	9.28	0.060	0.109	43.2	2027-11-04	
6-RA-103-02	10.97	4.57	2022-09-16	7.97	7.33	0.131	0.254	10.9	2027-11-04	
6-RA-103-03	10.97	4.57	2022-09-16	8.95	8.44	0.089	0.208	18.6	2027-11-04	
6-RA-103-04	10.97	4.57	2022-09-16	7.30	6.60	0.153	0.321	6.3	2027-11-04	gauge photo attached
6-RA-103-05	10.97	4.57	2022-09-16	7.43	6.91	0.141	0.281	8.3	2027-11-04	gauge photo attached
6-RA-103-06	10.97	4.57	2022-09-16	9.45	9.16	0.065	0.125	36.8	2027-11-04	
6-RA-103-07	10.97	4.57	2022-09-16	9.15	8.89	0.073	0.139	31.2	2027-11-04	
6-RA-103-08	10.97	4.57	2022-09-16	9.29	9.07	0.066	0.131	34.4	2027-11-04	
6-RA-103-09	10.97	4.57	2022-09-16	9.43	9.10	0.067	0.135	33.6	2027-11-04	
6-RA-103-10	10.97	4.57	2022-09-16	7.68	7.00	0.140	0.271	9.0	2027-11-04	gauge photo attached
6-RA-103-11	10.97	4.57	2019-10-18	8.91	5.92	0.413	0.592	2.3	2025-12-25	gauge photo attached
6-RA-103-12	10.97	4.57	2022-09-16	9.11	8.74	0.154	0.154	27.0	2027-11-04	
6-RA-103-13	10.97	4.57	2022-09-16	9.13	8.92	0.112	0.125	34.9	2027-11-04	
6-RA-103-14	10.97	4.57	2022-09-16	7.34	6.52	0.345	0.345	5.7	2027-09-02	gauge photo attached

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2022-09-16	5.92	5.74	0.065	0.061	33.8	2027-11-04	
3-RV-101-02	7.62	3.56	2022-09-16	5.81	5.70	0.065	0.053	32.8	2027-11-04	
3-RV-101-03	7.62	3.56	2022-09-16	4.98	4.72	0.098	0.103	11.3	2027-11-04	
3-RV-101-04	7.62	3.56	2022-09-16	5.39	5.18	0.084	0.079	19.3	2027-11-04	
3-RV-101-05	7.62	3.56	2022-09-16	5.91	5.84	0.064	0.053	35.7	2027-11-04	
3-RV-101-06	7.62	3.56	2022-09-16	5.39	5.22	0.083	0.065	20.1	2027-11-04	
3-RV-101-07	7.62	3.56	2022-09-16	4.80	4.63	0.111	0.087	9.6	2027-11-04	gauge photo attached
3-RV-101-08	7.62	3.56	2022-09-16	5.81	5.64	0.076	0.044	27.5	2027-11-04	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2022-09-16	3.91	3.76	0.062	0.079	12.2	2027-11-04	
2-RV-102-02	5.54	2.79	2022-09-16	3.64	3.49	0.073	0.073	9.6	2027-11-04	gauge photo attached
2-RV-102-03	5.54	2.79	2022-09-16	4.34	4.23	0.044	0.028	33.0	2027-11-04	
2-RV-102-04	5.54	2.79	2022-09-16	4.47	4.39	0.042	0.040	38.5	2027-11-04	
2-RV-102-05	5.54	2.79	2022-09-16	4.59	4.57	0.033	0.042	42.8	2027-11-04	
2-RV-102-06	5.54	2.79	2022-09-16	4.05	3.94	0.062	0.065	17.6	2027-11-04	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2019-10-18	5.40	5.15	0.012	0.050	>50	2030-11-04	
3-RF-101-02	5.49	2.41	2019-10-18	5.38	5.24	0.007	0.028	>50	2030-11-04	
3-RF-101-03	5.49	2.41	2019-10-18	5.35	5.35	0.003	0.000	>50	2030-11-04	
3-RF-101-04	5.49	2.41	2019-10-18	5.29	5.21	0.009	0.016	>50	2030-11-04	
3-RF-101-05	5.49	2.41	2019-10-18	5.17	5.14	0.011	0.006	>50	2030-11-04	
3-RF-101-06	5.49	2.41	2019-10-18	5.12	5.13	0.014	-0.002	>50	2030-11-04	
3-RF-101-07	5.49	2.41	2019-10-18	5.38	5.25	0.006	0.026	>50	2030-11-04	
3-RF-101-08	5.49	2.41	2019-10-18	5.23	5.12	0.022	0.022	>50	2030-11-04	
3-RF-101-09	5.49	2.41	2019-10-18	5.13	5.10	0.002	0.006	>50	2030-11-04	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2019-10-18	3.66	3.55	0.013	0.022	>50	2030-11-04	
2-RF-102-02	3.91	2.16	2019-10-18	3.65	3.53	0.012	0.024	>50	2030-11-04	
2-RF-102-03	3.91	2.16	2019-10-18	3.68	3.65	0.010	0.006	>50	2030-11-04	
2-RF-102-04	3.91	2.16	2019-10-18	2.83	1.94	0.067	0.176	expired	immediate	< t-min; gauge photo attached
2-RF-102-05	3.91	2.16	2019-10-18	2.88	1.92	0.159	0.190	expired	immediate	< t-min; gauge photo attached
2-RF-102-06	3.91	2.16	2019-10-18	2.74	1.92	0.160	0.162	expired	immediate	< t-min; gauge photo attached
2-RF-102-07	3.91	2.16	2019-10-18	2.82	1.86	0.073	0.190	expired	immediate	< t-min; gauge photo attached

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	2019-10-18	5.16	5.02	0.037	0.028	>50	2030-11-04	
4-LA-104-02	6.02	3.18	2019-10-18	4.79	4.55	0.052	0.048	26.5	2030-11-04	
4-LA-104-03	6.02	3.18	2019-10-18	5.23	5.20	0.027	0.006	>50	2030-11-04	
4-LA-104-04	6.02	3.18	2019-10-18	5.49	5.34	0.025	0.030	>50	2030-11-04	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-05	6.02	3.18	2019-10-18	4.71	4.37	0.058	0.067	17.7	2030-11-04	
4-LA-104-06	6.02	3.18	2019-10-18	5.38	5.17	0.032	0.042	47.8	2030-11-04	
4-LA-104-07	6.02	3.18	2019-10-18	5.15	4.92	0.038	0.046	38.2	2030-11-04	
4-LA-104-08	6.02	3.18	2019-10-18	4.88	4.63	0.046	0.050	29.3	2030-11-04	
4-LA-104-09	6.02	3.18	2019-10-18	4.61	4.37	0.048	0.048	25.0	2030-11-04	
4-LA-104-10	6.02	3.18	2019-10-18	4.63	4.32	0.060	0.061	18.6	2030-11-04	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2022-09-16	6.61	6.33	0.076	0.162	15.5	2027-11-04	
4-RA-104-02	8.56	3.81	2022-09-16	7.04	6.84	0.060	0.093	32.5	2027-11-04	
4-RA-104-03	8.56	3.81	2022-09-16	7.70	7.67	0.065	0.053	>50	2027-11-04	
4-RA-104-04	8.56	3.81	2022-09-16	7.31	6.98	0.109	0.131	24.2	2027-11-04	
4-RA-104-05	8.56	3.81	2022-09-16	7.12	6.74	0.119	0.164	17.8	2027-11-04	
4-RA-104-06	8.56	3.81	2022-09-16	6.70	6.41	0.076	0.143	18.2	2027-11-04	
4-RA-104-07	8.56	3.81	2022-09-16	6.04	5.61	0.104	0.210	8.6	2027-11-04	gauge photo attached
4-RA-104-08	8.56	3.81	2022-09-16	7.09	6.90	0.093	0.111	27.9	2027-11-04	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	2019-10-18	3.59	3.42	0.018	0.034	33.6	2030-11-04	
2-ST-101-02	3.91	2.29	2019-10-18	3.34	3.19	0.025	0.030	30.3	2030-11-04	
2-ST-101-03	3.91	2.29	2019-10-18	3.68	—	—	—	—	—	inaccessible – no reading
2-ST-101-04	3.91	2.29	2019-10-18	3.52	3.50	0.011	0.004	>50	2030-11-04	
2-ST-101-05	3.91	2.29	2019-10-18	3.27	3.22	0.024	0.010	38.6	2030-11-04	
2-ST-101-06	3.91	2.29	2019-10-18	3.66	3.66	-0.003	0.000	>50	2030-11-04	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	2019-10-18	4.91	4.79	0.022	0.024	>50	2030-11-04	
3-CW-101-02	5.49	2.79	2019-10-18	4.98	4.69	0.025	0.057	33.1	2030-11-04	
3-CW-101-03	5.49	2.79	2019-10-18	4.87	4.63	0.031	0.048	38.7	2030-11-04	
3-CW-101-04	5.49	2.79	2019-10-18	5.20	5.22	0.008	-0.004	>50	2030-11-04	
3-CW-101-05	5.49	2.79	2019-10-18	5.14	5.19	0.010	-0.010	>50	2030-11-04	
3-CW-101-06	5.49	2.79	2019-10-18	5.01	4.91	0.019	0.020	>50	2030-11-04	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	2019-10-18	4.85	4.77	0.025	0.016	>50	2030-11-04	
3-CW-102-02	5.49	2.79	2019-10-18	4.17	4.09	0.050	0.016	26.2	2030-11-04	
3-CW-102-03	5.49	2.79	2019-10-18	4.27	4.12	0.049	0.030	27.0	2030-11-04	
3-CW-102-04	5.49	2.79	2019-10-18	4.50	4.29	0.043	0.042	35.2	2030-11-04	
3-CW-102-05	5.49	2.79	2019-10-18	4.45	4.35	0.036	0.020	43.9	2030-11-04	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT/VT Level II; CAN/CGSB-48.9712 UT Level II	2024-11-07	_____